07 December 2025 13:36

Java → object oriented Language

↓

primitive data types → non objects

↓

int, char, boolean, float

To overcome this

↓

wrapper classes :- they are used to ~~wrap~~ encapsulate primitive data types.

primitive data types :- wrapper class :- object

primitive ⇄ wrapper class

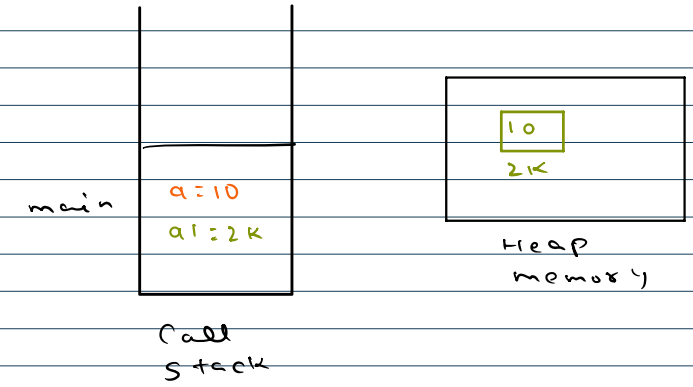
Primitive	Wrapper
int	Integer
byte	Byte
short	Short
long	Long
float	Float
double	Double
char	Character
boolean	Boolean

unboxing

autoboxing

```
public class WrapperClassesDemo {
    public static void main(String args[]) {
        int a = 10;
        Integer a1 = 10; //auto-boxing

        System.out.println("Hello Akarsh!");
    }
}
```



Cache Range

Integer, Byte, Short, Long :- -128 to 127

Character :- 0 to 127

Boolean :- true/false

Float, Double :- Never in cache, always new instance

Boolean => true/false

Float, Double => never in cache, always new instance

Applicable → autoboxing
→ gradeof
→ ~~new~~ → x (not applicable)

Array List

It is a resizable array and its implementation is provided java.

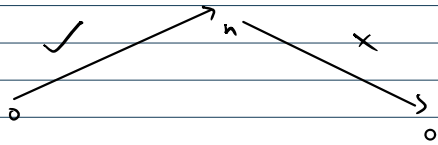
Functionalities → does not have a fixed size → grow dynamically
→ shrink dynamically

ArrayList → class → object
↓
Non-primitive data type
↓
Heap memory

Features → Indexing
→ add, remove, search, sort
→ Generic => String, Integer, Float
↓
Type → Wrapper classes
→ User defined class

For each loop

Reverse → x



new cap = old cap + (old/2) + 1. => 1.5 x

10 => 16 (10 + (10/2) + 1) = (10 + 5 + 1) = 16

Print all substrings length wise

akash

Print all substrings length wise

akarsh

len 1 => a, k, a, r, s, h
len 2 => ak, ka, ar, rs, sh
len 3 => aka, kar, ars, rsh
len 4 => akar, kars, arsh
len 5 => akars, karsh
len 6 => akarsh

0 1 2 3 4 5
a k a r s h

i j

0-1 a
1-2 k
2-3 a
3-4 r
4-5 s
5-6 h
j-i=1 (len)

0-2 ak
1-3 ka
2-4 ar
3-5 rs
4-6 sh
j-i=2 (len)

0-3 aka
1-4 kar
2-5 ars
3-6 rsh
j-i=3 (len)

0-4 akar
1-5 kars
2-6 arsh
j-i=4 (len)

0-5 akars
1-6 karsh
j-i=5 (len)

0-6 akarsh
j-i=6 (len)

j-i=len
j-len=i
i=j-len

```
for(len=1; len<=s.length(); len++) {
    for(int j=len; j<=s.length(); j++) {
        int i=j-len;
        sop(substring(i,j));
    }
}
```